

Clarification for the debt and temperature data type

“Debt” is a **ratio-scaled attribute**.

Reasoning:

- A debt ratio has a **true zero** (0 means no debt relative to the base, e.g., assets or income).
- Ratios are **meaningfully comparable**: a debt ratio of 0.6 indicates twice the debt burden of 0.3, all else equal.
- All arithmetic operations (addition, subtraction, multiplication, division) are meaningful.

Contrast with interval scale:

- Interval scales lack a true zero (e.g., temperature in Celsius).
- Ratios between values on an interval scale are not meaningful.

Conclusion:

Debt ratio satisfies the defining properties of a **ratio scale**, not an interval scale.

Temperature

Absolute zero vs. scale zero

- **Absolute zero** is a *physical concept* (0 Kelvin = no thermal motion).
- **Scale zero** is a *property of the measurement scale being used*.

In measurement theory, what matters is the **zero point of the chosen scale**, not whether a physical absolute zero exists in nature.

Why Celsius is an interval scale

- The zero in **Celsius (0 °C)** is **arbitrary** (freezing point of water).
- Because the zero is arbitrary:
 - Differences are meaningful: $20\text{ °C} - 10\text{ °C} = 10\text{ °C}$.
 - Ratios are **not** meaningful: 20 °C is *not* “twice as hot” as 10 °C .

Hence, Celsius is an **interval scale**.

Why Kelvin is a ratio scale

- **Kelvin** sets its zero at **absolute zero**.

- Zero is **non-arbitrary** and indicates total absence of thermal energy.
- Ratios are meaningful:
- 200 K **is** twice the thermal energy of 100 K (approximately, in thermodynamic terms).

Thus, Kelvin is a **ratio scale**.

Key takeaway

- A variable can have a *physical* absolute zero, yet still be **interval-scaled** if the measurement scale does not use that zero.
- **Ratio comparisons are valid only when the measurement scale has a true, non-arbitrary zero.**

So:

- Temperature in **Celsius or Fahrenheit** → interval scale
- Temperature in **Kelvin** → ratio scale